$= \sqrt{30 \times 2 \times 30 \times 2 \times 10 \times 10}$

$= 600 \ m^2$

In Fig.7.5, Area of triangle $= \sqrt{60(60-35)(60-40)(60-45)}$

$= \sqrt{60 \times 25 \times 20 \times 15}$

$= \sqrt{20 \times 3 \times 5 \times 5 \times 20 \times 3 \times 5}$

$= 300\sqrt{5}$ (Since $\sqrt{5} = 2.236$)

$= 670.8 \ m^2$

In Fig.7.6, Area of triangle $= \sqrt{60(60-40)(60-40)(60-40)}$

$= \sqrt{60 \times 20 \times 20 \times 20}$

$= \sqrt{3 \times 20 \times 20 \times 20 \times 20}$

$= 400\sqrt{3}$ (Since $\sqrt{3} = 1.732$)

$= 692.8 \ m^2$

We find that though the perimeters are same, the areas of the three triangular plots are different. The area of the triangle in Fig 7.6 is the greatest among these; the buyer can be suggested to choose this since it is more spacious.

Note

If the perimeter of different types of triangles have the same value, among all the types of triangles, the equilateral triangle possess the greatest area. We will learn more about maximum areas in higher classes.

7.3 Application of Heron's Formula in Finding Areas of Quadrilaterals

A plane figure bounded by four line segments is called a quadrilateral.

Let $ABCD$ be a quadrilateral. To find the area of a quadrilateral, we divide the quadrilateral into two triangular parts and use Heron's formula to calculate the area of the triangular parts.

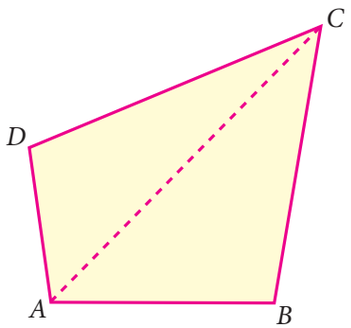


Fig. 7.7

In Fig 7.7,

Area of quadrilateral $ABCD$ = Area of triangle ABC + Area of triangle ACD

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